Replication package for: Indirect Variational Inference: Applications to Earnings Dynamics

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Overview

This package reproduces every estimate, figure and table of *Indirect Variational Inference: Applications to Earnings Dynamics* by Neele Balke, Stéphane Bonhomme and Thibaut Lamadon (Journal of Labor Economics; arXiv:2607.15168). The code is in Python (PyTorch for estimation, numpy and Tectonic for figures and tables).

The paper has two parts:

- **Monte Carlo simulations** (Tables 1–3, Figures 1–8, web-appendix Tables E1 and F1). These need no external data.
- **An application to PSID earnings** (Table 4, Figures 9–12, Appendix Table B1, web-appendix Table H1). It uses the PSID household panel prepared by Blundell, Pistaferri and Preston (2008), which replicators download themselves from the ICPSR–PSID Data Extract Repository (ICPSR 210782). This package contains no PSID data.

The pipeline has three layers, all driven by `replicate.py` (or by the one-command wrapper `run_all.sh`):

Layer	What it does	Output	Time
1: data	Builds the PSID earnings matrix and sample statistics from the ICPSR files	`output/data/`	minutes (CPU)
2: bundles	Runs the estimations and simulations and collects results into JSON *bundles*	`output/bundles/`	~22 GPU-hours (+ ~210 for the optional bootstrap)
3: figures	Builds every figure and table from the bundles	`output/figures/`	~2 minutes (CPU)

The bundles produced by the authors' run are included, so **layer 3 can be run on its own** in about two minutes, without the data and without a GPU, to rebuild all figures and tables from the published estimates.

Data Availability and Provenance Statements

Statement about rights

- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ☒ I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package. The package contains only simulated data, estimation results and aggregate sample statistics; it contains no PSID microdata.

License for data

The whole package, including the results in `output/`, is licensed under the MIT License; see `LICENSE`. The PSID-derived input files are not part of the package and are governed by the ICPSR and PSID terms of use.

Summary of availability

- ☐ All data **are** publicly available.
- ☒ Some data **cannot be made** publicly available.
- ☐ **No data can be made** publicly available.

The simulation results use data generated by the code. The empirical application uses PSID data that cannot be redistributed, but which any researcher can obtain free of charge as described below.

Details on each data source

Data source	Files	Provided	Access
Blundell, Pistaferri and Preston (2008) PSID extract, ICPSR 210782 V2	`data.dta`, `tax9192.dta`, `natpr.dta`	No	Free ICPSR account; accept the ICPSR and PSID terms of use
Simulated panels	generated by `scripts/_simulation_*.py`, `_time_to_convergence_hockey.py`, `_contour_past_2period_imputed.py`	Regenerated by the code	–

PSID data (Blundell, Pistaferri and Preston 2008). The empirical application starts from the household-year panel that Blundell, Pistaferri and Preston (BPP) built from the Panel Study of Income Dynamics (PSID) for "Consumption Inequality and Partial Insurance" (American Economic Review, 2008). Their replication package is deposited in the ICPSR–PSID Data Extract Repository as study 210782 (version V2, doi:10.3886/E210782V2). We use three of its files:

File	Content
`data.dta`	BPP's PSID household-year panel, built by their `create1_AER.do` and `create2_AER.do` from the PSID family files (1968–1992) and the 1968–97 cross-year individual file

File	Content
`tax9192.dta`	TAXSIM federal income taxes for 1991–92
`natpr.dta`	National price index (`cpi`)

PSID Conditions of Use prohibit redistributing PSID data, so these files and everything derived from them at the individual level (in particular the 741 × 10 earnings matrix `output/data/bpp_y_matrix.npy`) are not included. To obtain them:

1. Create a free account at ICPSR and register with the PSID, accepting the PSID Conditions of Use.
2. Download the files from https://www.icpsr.umich.edu/sites/psid/view/studies/210782/versions/V2 (the data files are in BPP's package; we use only the three above).
3. Extract and verify them in one step: `python3 scripts/extract_icpsr_inputs.py PATH/TO/the-downloaded.zip` writes the three files to `data/raw/` and checks their checksums. (Alternatively, place them in `data/raw/` yourself, or set `BPP_EXT` to their directory.)
4. The expected SHA-256 checksums (`sha256sum data/raw/*.dta`) are:

```
c1d1949748e224f03b80803b01ea7775cb043cae1da5e89a8717452591ecfd41  data.dta
0af8a9b32b1b6d3c3d80e2f0830f662bf82c23f1d51a3c27d601f07efb17af81  tax9192.dta
7044d5511d7a3caa079688fe3513043907162373403f7dd53bfeaf1fa2a4cd9b  natpr.dta
```

`scripts/psid_bpp_build.py` ports BPP's sample selection (`adjust_AER.do`), the PSID part of `impute_AER.do` and the income definition in `mindist_AER.do` to Python, restricts to 1980–1989 and the balanced panel, and residualizes log earnings. After layer 1, `uv run python replicate.py verify-data` checks that the resulting matrix has the expected SHA-256 (`c9cda89925b3c35ae67ebaac8a3dfb748ac55d76750d609c0879d61555a9ffae`).

Data citations: Blundell, Pistaferri and Preston (2025) for the extract and Panel Study of Income Dynamics (2026) for the underlying survey; see References.

Dataset list

File	Source	Notes	Provided
`data/raw/data.dta`	ICPSR 210782 V2	PSID household-year panel (BPP)	No
`data/raw/tax9192.dta`	ICPSR 210782 V2	TAXSIM taxes 1991–92	No
`data/raw/natpr.dta`	ICPSR 210782 V2	Price index	No
`output/data/bpp_y_matrix.npy`	layer 1	Residual log earnings, 741 households × 10 years (PSID microdata)	No
`output/data/bpp_selection_table.json`	layer 1	Aggregate sample statistics (counts, means, moments)	Yes
`output/bundles/*.json` (11 files)	layer 2	Estimates, simulation results, bootstrap summaries	Yes
`output/bundles/cells/_bpp_hetero_scale_ivi_bootstrap_k40*/bootstrap_theta.json`	layer 2	Bootstrap replicates of the parameter vector	Yes

The included files contain parameter estimates and aggregate statistics only. Each bundle is described in Appendix A.

Computational requirements

Software requirements

- Linux x86-64 (tested on NixOS and RHEL 8). Other platforms are untested.
- mise, which installs the pinned tools from `mise.toml`:
  - uv 0.12.0 (Python environment manager)
  - Tectonic 0.17.0 (self-contained LaTeX engine for layer 3; downloads about 50 MB of packages from its versioned TeX bundle on first use)
- Python 3.12, with all packages pinned in `uv.lock` (`requirements.lock` has the same pins for pip). The main ones:
  - torch 2.13.0 (with CUDA 13 libraries)
  - numpy 2.5.2, scipy 1.18.1, pandas 3.0.5, statsmodels 0.14.6, matplotlib 3.11.1
  - pydantic, structlog, nvidia-ml-py; pytest for the checks
- Ghostscript ≥ 10, only for the journal EPS files (`gs` on `PATH`).
- No Stata, R, MATLAB or full TeX distribution is needed.

`mise run setup` (or `uv sync --frozen`) creates the environment, about 5 GB. On NixOS, PyTorch needs `/run/opengl-driver/lib` and 64-bit `libstdc++` and `libz` on `LD_LIBRARY_PATH`: `nix develop` (from `flake.nix`) supplies the driver path and `libstdc++`; also make `libz` available, and set `TRITON_LIBCUDA_PATH=/run/opengl-driver/lib`.

Controlled randomness

☒ Random seeds are set at the start of every stochastic program.

Every stochastic entry point calls `mlye.seeding.seed_everything` before any draw (`tests/test_replication.py` enforces this). It seeds Python, numpy and torch (CPU and CUDA), turns on `torch.use_deterministic_algorithms`, disables cuDNN benchmarking and sets `CUBLAS_WORKSPACE_CONFIG`. Each script then uses explicit seeds, all passed by `replicate.py`, which also sets `PYTHONHASHSEED=0`:

- simulated panel: `OBS_SEED=11`
- common-random-number draws: `NOISE_SEED=12345`
- encoder initialization: `VI_SEED=11007`; NaN-retry sequence `VI_SEED + i`
- PSID fits: `*_SEED=11`
- bootstrap: household resample seed `31000+b`; the unconditional bootstrap also uses `SEED=5000+b` and `NOISE_SEED=900000+b`

Rerunning on the same GPU model, driver and torch build gives bitwise-identical outputs. Across GPU models (the original runs used H100 and L40S GPUs), floating-point reduction order differs, so results agree to optimization precision. `replicate.py compare` quantifies the differences. The wall-clock times in Table E1 depend on the hardware.

Memory, runtime and storage requirements

Stage	Hardware	Time measured in the authors' run
Layer 1: data	CPU	Minutes
Layer 2: estimations and simulations	NVIDIA H100 / L40S GPUs	22 GPU-hours in total (14.5 h H100, 7.7 h L40S); allow 1–2 days sequentially on one GPU
Optional bootstrap (2 × 100 IVI replicates)	NVIDIA H100	About 210 GPU-hours (estimate)
Layer 3: figures and tables	CPU / laptop	About 2 minutes

- **GPU:** NVIDIA GPU with at least 4 GB of memory and a CUDA 13 compatible driver (≥ 580). Measured peak GPU memory was 2.3 GB. Scripts fall back to the CPU, but layer 2 is then impractically slow.
- **Memory:** 16 GB of RAM (cluster jobs requested 32 GB).

- **Storage:** about 6 GB (5 GB for the Python environment).
- **Authors' hardware:** the `pythia` Slurm cluster (NVIDIA H100 and L40S nodes, RHEL 8) for layers 1–2; a NixOS workstation (NVIDIA RTX 5090) and GitHub Actions (`ubuntu-latest`) for layer 3.

The longest task, the T = 40 hockey-stick IVI fit, took 10.5 hours on an H100. The run summary below is generated from the run's provenance manifests.

Run summary

Measured on the replication run of 2026-10-05 (commit `08c6169`, `759ab56`, 206 tasks with manifests). GPU-hours are task wall-clock time on the GPU type the task actually ran on.

Dependency graph *The dependency graph (data → tasks → bundles → figures) is rendered in README.md on GitHub; the tables below give the same information.*

Compute by block

Block	Tasks	GPU-hours	Longest task	Elapsed (first start → last finish)
data	1	CPU only	`bpp-sample-stats` (0.0 h)	0.0 h
ar1n	157	2.8 h L40S	`ar1n-sweep-anderson_meanfield_alpha060_m4_ep8000_lr1e-2_crn` (0.2 h)	1.0 h
hetero	7	1.1 h L40S	`hetero-anderson_alpha060_m4_ep16000_lr1e-2_crn_rho030` (0.4 h)	0.4 h
hockey	15	10.5 h H100, 2.7 h L40S	`hockey-t40-picard_cold` (10.5 h)	10.7 h
timing	17	1.1 h L40S	`timing-smc-big-K200` (0.2 h)	0.6 h
bpp	9	4.0 h H100	`bpp-hs-ivi-it50` (2.9 h)	3.1 h
total	**206**	**14.5 h H100, 7.7 h L40S** (= 22 GPU-hours)		
bootstrap (not regenerated)	200 + 2	~210 h H100 (estimate)		

The data matrix (`bpp-data`, in the bpp block) and the sample statistics (`data` block) run on a CPU in minutes; so does layer 3 (figures).

Bundles and where they are used

Bundle	Block	Tasks in lineage	GPU-hours in lineage	Used by
`ar1n_parameter_summary.json`	ar1n	14	0.9 h L40S	Table 1 (`ar1n_parameter_table`)

Bundle	Block	Tasks in lineage	GPU-hours in lineage	Used by
`ar1n_mu1_profile_bundle.json`	ar1n	23	0.5 h L40S	Figure 4 (`ar1n_likelihood_elbo`)
`ar1n_binding_mu1_bundle.json`	ar1n	19	0.4 h L40S	Figure 5 (`ar1n_binding_function`)
`ar1n_quiver_bundle.json`	ar1n	104	1.5 h L40S	Figure 3 (`ar1n_trajectory_fixed_point`)
`hetero_scale_parameter_summary.json`	hetero	7	1.1 h L40S	Table 3 (`hetero_scale_parameter_table`), Figure 8 (`hetero_scale_a_sigma_z1_e`)
`hockeystick_parameter_summary.json`	hockey	12	2.7 h L40S	Table 2 (`hockeystick_parameter_table`), Figure 7 (`hockeystick_mu_sigma_z1_e`)
`hockeystick_parameter_summary_longt.json`	hockey	2	10.5 h H100	Table 2 (`hockeystick_parameter_table`)
`contour_bundle.json`	hockey	2	0.9 h L40S	Figure 6 (`hockeystick_posteriors`)
`time-to-convergence-hockey-bundle.json`	timing	17	1.1 h L40S	Table E1 (web appendix) (`hockeystick_resource_table`)
`bundle_psid_heteroscale.json`	bpp	6	3.3 h H100	Table 4 (`psid_parameter_table`), Figure 9 (`psid_a_sigma_z1_eps`), Figure 10 (`psid_persistence_residual`), Figure 11 (`psid_mixture_density`), Figure 12 (`psid_ce_risk_premium`), Table H1 (web appendix) (`psid_hetro_comp_parameter_table`)
`2026-07-13-bpp-hetero-mean-indep-bundle.json`	bpp	5	3.7 h H100	Table H1 (web appendix) (`psid_hetro_comp_parameter_table`)
`bpp_selection_table.json`	data	1	CPU only	Table B1 (`psid_sample_table`)

Lineages overlap (e.g. the ar1n Picard cell feeds four bundles), so the per-bundle hours do not add up to the block totals.

Figures and tables

Artifact	Generator	Bundles	Bootstrap
Figure 1	scripts/fig_sample_vs_population.py	fixed content	—
Figure 2	scripts/fig_ivi_binding_function.py	fixed content	—
Table 1	scripts/fig_ar1n_parameter_table.py	ar1n_parameter_summary.json	—
Figure 3	scripts/fig_ar1n_trajectory_fixed_point.py	ar1n_quiver_bundle.json	—
Figure 4	scripts/fig_ar1n_likelihood_elbo.py	ar1n_mu1_profile_bundle.json	—
Figure 5	scripts/fig_ar1n_binding_function.py	ar1n_binding_mu1_bundle.json	—
Table 2	scripts/fig_hockeystick_parameter_table.py	hockeystick_parameter_summary.json, hockeystick_parameter_summary_longt.json	—
Figure 7	scripts/fig_hockeystick_mu_sigma_z1_e.py	hockeystick_parameter_summary.json	—
Figure 6	scripts/fig_hockeystick_posteriors.py	contour_bundle.json	—
Table E1 (web appendix)	scripts/fig_hockeystick_resource_table.py	time-to-convergence-hockey-bundle.json	—
Table 3	scripts/fig_hetero_scale_parameter_table.py	hetero_scale_parameter_summary.json	—
Figure 8	scripts/fig_hetero_scale_a_sigma_z1_e.py	hetero_scale_parameter_summary.json	—
Table 4	scripts/fig_psid_parameter_table.py	bundle_psid_heteroscale.json	IVI-row standard errors (100 replicates)
Figure 9	scripts/fig_psid_a_sigma_z1_eps.py	bundle_psid_heteroscale.json	95% band around the IVI curves (100 replicates)
Figure 10	scripts/fig_psid_persistence_residual.py	bundle_psid_heteroscale.json	—
Figure 11	scripts/fig_psid_mixture_density.py	bundle_psid_heteroscale.json	—
Figure 12	scripts/fig_psid_ce_risk_premium.py	bundle_psid_heteroscale.json	—
Table H1 (web appendix)	scripts/fig_psid_hetro_comp_parameter_table.py	2026-07-13-bpp-hetero-mean-indep-bundle.json, bundle_psid_heteroscale.json	—
Table B1	scripts/fig_psid_sample_table.py	bpp_selection_table.json	—
Table F1 (web appendix)	scripts/fig_computational_settings_table.py	fixed content	—

The bootstrap quantities come from the shipped `output/bundles/cells/_bpp_hetero_scale_ivi_bootstrap_k40_uncond/bootstrap_theta.json` until the bootstrap stage is rerun (`hut.replication_bpp_with_bootstrap.json`).

Description of programs

Path	Purpose
`run_all.sh`	One-command replication (layers 1 → 2 → 3, data check, comparison with the published bundles)
`replicate.py`	Single source of truth for the pipeline: every task, its layer, settings, seeds, dependencies and resources, and the registry of figures and tables (`FIGURES`)
`scripts/psid_bpp_*.py`	Layer 1: PSID earnings matrix and sample statistics
`scripts/bpp_*.py`	Layer 2: estimations on the PSID
`scripts/_simulation_*.py`, `_time_to_convergence_hockey.py`, `_contour_*.py`	Layer 2: simulation experiments
`scripts/_export_*.py`, `*_aggregate.py`, `bpp_hetero_scale_bundle.py`	Layer 2: assemble the bundles
`scripts/fig_*.py`	Layer 3: one generator per figure or table, plus compilation (`fig_compile.py`) and EPS export (`fig_eps.py`)
`mlye/`	Estimation library: encoders, decoders, priors, the full VI model, SMC and FIVO
`tests/`	Consistency checks (pipeline graph, imports, seeding, environment pins, figure registry)

Every file is listed in Appendix A. A few provenance fields in the bundles name entries of the authors' research log (`journal/…`); the log is not part of the package and is not needed to run it.

License for code

The code (like the rest of the package) is licensed under the MIT License; see `LICENSE`.

Instructions to replicators

1. **Install the tools.** On Linux x86-64 with git, install mise, then from the package root run:

   ```
   mise install       # uv 0.12.0 and Tectonic 0.17.0
   mise run setup     # Python 3.12 + pinned packages (uv sync --frozen)
   mise run test      # consistency checks, about 1 minute
   ```

   Without mise, install uv and Tectonic yourself and run `uv sync --frozen` and `uv run pytest`. Internet access is needed for this step.

2. **Rebuild the figures and tables from the published estimates** (no data, no GPU, about 2 minutes):

   ```
   ./run_all.sh --figures-only
   ```

   This writes `output/figures/figures.pdf` (all figures and tables, labelled with their paper numbers), one PDF and one LaTeX fragment per item, and journal EPS files.

3. **Obtain the PSID data** as described in Details on each data source, then run `python3 scripts/extract_icpsr_inputs.py PATH/TO/the-downloaded.zip`.

4. **Run the full replication:**

   ```
   ./run_all.sh                   # layers 1-3, using the published bootstrap
   ./run_all.sh --with-bootstrap  # also rerun the 2 x 100 bootstrap
   ```

   `run_all.sh` builds the data (layer 1), checks the earnings matrix checksum, reruns every estimation and simulation (layer 2), rebuilds the figures and tables from the new bundles (layer 3),

and compares the new bundles with the published ones. Regenerating overwrites the files in `output/`; `git restore output/` brings back the published versions.

Details

- **Partial runs.** `uv run python replicate.py run --layer 2 --block hetero` runs one block; `--only 'ar1n-sweep-*'` selects tasks by id; `--workdir DIR` writes to a separate tree; `uv run python replicate.py list` prints every task. The simulation blocks (`ar1n`, `hetero`, `hockey`, `timing`) need no data.
- **Bootstrap.** By default, the PSID standard errors and the confidence band in Figure 9 use the included bootstrap results (`output/bundles/cells/_bpp_hetero_scale_ivi_bootstrap_k40*/`). Add `--with-bootstrap` to recompute them.
- **Smoke test.** `uv run python replicate.py run --smoke --layer 1,2 --workdir /tmp/smoke` runs every task with tiny settings in minutes. It checks the code paths, not the results.
- **Colab.** `notebooks/table2_vi_gaussian.ipynb` reruns one line of Table 2 (the T = 6 unrestricted Gaussian VI fit) on a Colab GPU and compares it with the published estimates.
- **Cluster.** The authors ran layers 1–2 on a Slurm cluster through scripthut; see Appendix B.

List of tables, figures and programs

The provided code reproduces:

☐ All numbers provided in text in the paper (not yet checked against the final text)
☒ All tables and figures in the paper

Each item is generated by `scripts/fig_<name>.py` into `output/figures/tex/<name>.tex` (LaTeX fragment), compiled to `output/figures/pdf/<name>.pdf`; paper items are also exported to `output/figures/eps/<number>_<name>.eps`. All are collected in `output/figures/figures.pdf`.

Figure/Table	Program	Output file (`output/figures/…`)	Input bundle(s)
Figure 1	`scripts/fig_sample_vs_population.py`	`pdf/sample_vs_population.pdf`, `eps/fig1_sample_vs_population.eps`	fixed content
Figure 2	`scripts/fig_ivi_binding_function.py`	`pdf/ivi_binding_function.pdf`, `eps/fig2_ivi_binding_function.eps`	fixed content
Table 1	`scripts/fig_ar1n_parameter_table.py`	`pdf/ar1n_parameter_table.pdf`, `eps/tab1_ar1n_parameter_table.eps`	`ar1n_parameter_summary.json`
Figure 3	`scripts/fig_ar1n_trajectory_fixed_point.py`	`pdf/ar1n_trajectory_fixed_point.pdf`, `eps/fig3_ar1n_trajectory_fixed_point.eps`	`ar1n_quiver_bundle.json`
Figure 4	`scripts/fig_ar1n_likelihood_elbo.py`	`pdf/ar1n_likelihood_elbo.pdf`, `eps/fig4_ar1n_likelihood_elbo.eps`	`ar1n_mu1_profile_bundle.json`
Figure 5	`scripts/fig_ar1n_binding_function.py`	`pdf/ar1n_binding_function.pdf`, `eps/fig5_ar1n_binding_function.eps`	`ar1n_binding_mu1_bundle.json`
Table 2	`scripts/fig_hockeystick_parameter_table.py`	`pdf/hockeystick_parameter_table.pdf`, `eps/tab2_hockeystick_parameter_table.eps`	`hockeystick_parameter_summary.json`, `hockeystick_parameter_summary_longt.json`
Figure 6	`scripts/fig_hockeystick_posteriors.py`	`pdf/hockeystick_posteriors.pdf`, `eps/fig6_hockeystick_posteriors.eps`	`contour_bundle.json`

Figure/Table	Program	Output file (output/figures/…)	Input bundle(s)
Figure 7	scripts/fig_hockeystick_mu_sigma_z1_e.py	pdf/hockeystick_mu_sigma_z1_e.pdf, eps/fig7_hockeystick_mu_sigma_z1_e.eps	hockeystick_parameter_summary.json
Table 3	scripts/fig_hetero_scale_parameter_table.py	pdf/hetero_scale_parameter_table.pdf, eps/tab3_hetero_scale_parameter_table.eps	hetero_scale_parameter_summary.json
Figure 8	scripts/fig_hetero_scale_a_sigma_z1_e.py	pdf/hetero_scale_a_sigma_z1_e.pdf, eps/fig8_hetero_scale_a_sigma_z1_e.eps	hetero_scale_parameter_summary.json
Table 4	scripts/fig_psid_parameter_table.py	pdf/psid_parameter_table.pdf, eps/tab4_psid_parameter_table.eps	bundle_psid_heteroscale.json
Figure 9	scripts/fig_psid_a_sigma_z1_eps.py	pdf/psid_a_sigma_z1_eps.pdf, eps/fig9_psid_a_sigma_z1_eps.eps	bundle_psid_heteroscale.json
Figure 10	scripts/fig_psid_persistence_residual.py	pdf/psid_persistence_residual.pdf, eps/fig10_psid_persistence_residual.eps	bundle_psid_heteroscale.json
Figure 11	scripts/fig_psid_mixture_density.py	pdf/psid_mixture_density.pdf, eps/fig11_psid_mixture_density.eps	bundle_psid_heteroscale.json
Figure 12	scripts/fig_psid_ce_risk_premium.py	pdf/psid_ce_risk_premium.pdf, eps/fig12_psid_ce_risk_premium.eps	bundle_psid_heteroscale.json
Table B1 (appendix)	scripts/fig_psid_sample_table.py	pdf/psid_sample_table.pdf, eps/tabB1_psid_sample_table.eps	bpp_selection_table.json
Table E1 (web appendix)	scripts/fig_hockeystick_resource_table.py	pdf/hockeystick_resource_table.pdf	time-to-convergence-hockey-bundle.json
Table F1 (web appendix)	scripts/fig_computational_settings_table.py	pdf/computational_settings_table.pdf	fixed content
Table H1 (web appendix)	scripts/fig_psid_hetro_comp_parameter_table.py	pdf/psid_hetro_comp_parameter_table.pdf	2026-07-13-bpp-hetero-mean-indep-bundle.json, bundle_psid_heteroscale.json

The bundles are in output/bundles/, except bpp_selection_table.json (output/data/). The run summary traces each bundle back to the layer-2 tasks that produce it. The PSID standard errors (Table 4) and the confidence band (Figure 9) use the bootstrap replicates in output/bundles/cells/_bpp_hetero_scale_ivi_bootstrap_k40_uncond/bootstrap_theta.json.

References

Balke, Neele, Stéphane Bonhomme, and Thibaut Lamadon. 2026. "Replication Data for: Indirect Variational Inference: Applications to Earnings Dynamics." Harvard Dataverse. https://doi.org/[DATAVERSE-DOI-PLACEHOLDER].

Blundell, Richard, Luigi Pistaferri, and Ian Preston. 2008. "Consumption Inequality and Partial Insurance." *American Economic Review* 98 (5): 1887–1921. https://doi.org/10.1257/aer.98.5.1887.

Blundell, Richard, Luigi Pistaferri, and Ian Preston. 2025. "Replication Data for: Consumption

Inequality and Partial Insurance." Version V2. ICPSR - Interuniversity Consortium for Political and Social Research (ICPSR–PSID Data Extract Repository). https://doi.org/10.3886/E210782V2.

Panel Study of Income Dynamics, public use dataset. Produced and distributed by the Survey Research Center, Institute for Social Research, University of Michigan, Ann Arbor, MI ([YEAR-OF-DOWNLOAD-PLACEHOLDER]).

Appendix A: File inventory

Every file in the package, except the per-figure tables above.

Top level

File	Purpose
`README.md`, `README.pdf`	This document (Markdown source and PDF)
`LICENSE`	MIT License (the whole package)
`run_all.sh`	One-command replication
`replicate.py`	Pipeline driver and figure registry
`pyproject.toml`	Python project and dependencies
`uv.lock`	Pinned Python environment (uv)
`requirements.lock`	The same pins for pip
`mise.toml`	Pinned tool versions (uv, Tectonic) and shortcut tasks
`flake.nix`, `flake.lock`	Nix development shell (NixOS only)
`scripthut.yaml`	Cluster environment and backend configuration for scripthut
`.github/workflows/release.yml`	GitHub Actions job building layer 3 and attaching `figures.pdf` to a release
`.gitignore`	Keeps data, environments and regenerated outputs out of git
`data/raw/README.md`	Where to put the ICPSR input files, with checksums
`notebooks/table2_vi_gaussian.ipynb`	Colab notebook rerunning one Table 2 estimation
`workflows/hut.replication_<block>[_smoke].json`	Generated scripthut workflows for layers 1–2 (12 files: blocks `ar1n`, `hetero`, `hockey`, `timing`, `bpp`, `bpp_with_bootstrap`, each full and smoke); regenerate with `replicate.py hut`

Layer 1: data (`scripts/`)

File	Purpose
`psid_bpp_build.py`	Port of BPP's sample selection; builds `output/data/bpp_y_matrix.npy`
`psid_bpp_selection_table.py`	Sample statistics: BPP as published vs our selection vs the balanced panel

Layer 2: estimation and simulation (`scripts/`)

File	Purpose
bpp_hetero_scale_vi_mf.py, bpp_hetero_scale_vi_jn.py, bpp_hetero_scale_vi_sm.py	VI fits of the heterogeneous-variance model on the PSID (mean-field, joint-normal, structured Markov posteriors)
bpp_hetero_scale_ivi_jn.py	IVI fit of the heterogeneous-variance model on the PSID; also each bootstrap replicate
bpp_hetero_mean_indep_vi_jn.py, bpp_hetero_mean_indep_ivi_jn.py	VI and IVI fits of the heterogeneous-mean model (Table H1)
_bpp_hetero_scale_ivi_bootstrap_aggregate.py	Aggregates the bootstrap replicates
bpp_hetero_scale_bundle.py	Builds bundle_psid_heteroscale.json
_export_bpp_hetero_mean_indep_bundle.py	Builds the heterogeneous-mean bundle
_simulation_ar1n_sweep.py, _ar1n_posterior_comparison.py	AR(1)+normal simulation fits (Table 1) and their model definitions
_simulation_ar1n_mu1_profile.py, _simulation_ar1n_binding_mu1.py, _simulation_ar1n_quiver.py	Likelihood/ELBO profile, binding function and quiver (Figures 3–5)
_export_ar1n_parameter_summary.py, _export_ar1n_mu1_profile_bundle.py, _export_ar1n_binding_mu1_bundle.py, _export_ar1n_quiver_bundle.py	AR(1) bundles
_simulation_hockeystick_ivi_sweep.py, _export_hockeystick_parameter_summary.py	Nonlinear (hockey-stick) model simulations (Table 2, Figure 7)
_contour_past_2period_imputed.py	Posterior contours (Figure 6)
_time_to_convergence_hockey.py, _export_time_to_convergence_hockey_bundle.py, gpu_util.py	VI vs SMC timing experiment (Table E1) and GPU-utilization sampler
_simulation_hetero_scale_ivi_sweep.py, _export_hetero_scale_parameter_summary.py	Heterogeneity-model simulations (Table 3, Figure 8)
_scale_hetero_correlation_sweep.py, _scale_hetero_correlation_ivi.py	Data-generating process and IVI helpers for the heterogeneity simulations
psid_smc_em_fivo_sinhz1.py, bpp_hockey_common.py	Shared model and training helpers for the PSID fits
smc_smoothers.py, smc_truth_profile.py, climb_per_step.py, softplus_sinh_posterior_comparison.py	Shared SMC smoothers and helpers imported by the experiments

Layer 3: figures and tables (scripts/)

File	Purpose
fig_<name>.py (20 files)	One generator per figure or table; see the table above
fig_compile.py	Compiles the fragments to PDF with Tectonic
fig_eps.py	Exports the paper items to EPS (Ghostscript)
fig_preamble.tex	The paper's LaTeX preamble, shared by all items
fig_parameter_names.md	Maps parameter symbols in the tables to bundle keys

Packaging (scripts/)

File	Purpose
readme_pdf.sh, readme/pdf.lua	Build README.pdf from README.md (pandoc, xelatex)
extract_icpsr_inputs.py	Extract the three ICPSR 210782 input files from the download into data/raw/ and verify their checksums

File	Purpose
`make_dataverse_bundle.sh`	Build the Dataverse zip from the committed tree and verify it contains no PSID data

Library (`mlye/`)

File	Purpose
`__init__.py`, `utils.py`	Package entry and utilities
`seeding.py`	`seed_everything`, called by every stochastic entry point
`models/full_model.py`	Full VI model: encoder, decoder and prior
`models/encoders/` (`base.py`, `normal.py`, `markov.py`, `laplace.py`, `layers.py`, `__init__.py`)	Variational posteriors q(z | y)
`models/decoders/` (`base.py`, `ma.py`, `__init__.py`)	Emission distributions p(y | z), including MA(1)
`models/priors/` (`base.py`, `ar.py`, `markov.py`, `utils.py`, `__init__.py`)	Priors: AR(1) and nonlinear Markov
`models/__init__.py`	Models package
`eval/` (`smc.py`, `fivo.py`, `__init__.py`)	Particle filter and FIVO for likelihood evaluation

Tests (`tests/`)

File	Purpose
`test_replication.py`	Pipeline graph, bundle coverage, seeding, environment pins, workflows, figure registry
`test_eval_sinh_arcsinh_normal.py`	Unit test of the sinh-arcsinh density
`conftest.py`, `__init__.py`	Pytest configuration

Estimation bundles

Bundle (`output/bundles/`)	Block	Contents	Used in
`ar1n_parameter_summary.json`	ar1n	AR(1)+normal simulation: 13 VI / IVI / MLE fits	Table 1
`ar1n_mu1_profile_bundle.json`	ar1n	ELBO/likelihood profile in μ_1 (20-point grid)	Figure 4
`ar1n_binding_mu1_bundle.json`	ar1n	IVI binding function in μ_1 (15-point grid)	Figure 5
`ar1n_quiver_bundle.json`	ar1n	Binding-function quiver on the (μ_1, log σ_ε) plane	Figure 3
`hockeystick_parameter_summary.json`	hockey	Nonlinear model simulation, T = 6	Table 2, Figure 7
`hockeystick_parameter_summary_longt.json`	hockey	Nonlinear model simulation, T = 40 IVI	Table 2
`contour_bundle.json`	hockey	Imputed-posterior contours	Figure 6
`time-to-convergence-hockey-bundle.json`	timing	VI vs SMC time to convergence	Table E1

Bundle (`output/bundles/`)	Block	Contents	Used in
`hetero_scale_parameter_summary.json`	hetero	Heterogeneity-model simulation: 6 fits	Table 3, Figure 8
`bundle_psid_heteroscale.json`	bpp	PSID heterogeneous-variance model: VI (3 posteriors), IVI, bootstrap	Table 4, Figures 9–12, Table H1
`2026-07-13-bpp-hetero-mean-indep-bundle.json`	bpp	PSID heterogeneous-mean model vs heterogeneous-variance model	Table H1
`cells/_bpp_hetero_scale_ivi_bootstrap_k40/bootstrap_theta.json`	bpp	Conditional bootstrap (100 replicates)	Table 4 (reference)
`cells/_bpp_hetero_scale_ivi_bootstrap_k40_uncond/bootstrap_theta.json`	bpp	Unconditional bootstrap (100 replicates)	Table 4, Figure 9
`output/data/bpp_selection_table.json`	data	PSID sample statistics	Table B1

Appendix B: Pipeline management

Published outputs and comparisons

The published outputs tracked in git are the 11 estimation bundles (`output/bundles/*.json`), the sample statistics (`output/data/bpp_selection_table.json`) and the two bootstrap aggregates (`output/bundles/cells/_bpp_hetero_scale_ivi_bootstrap_k40*/`). Everything else under `output/` is git-ignored and can be recreated.

Regeneration overwrites published files in place. Use `git diff output/` to inspect changes and `uv run python replicate.py compare` to quantify them against the published bundles at git `HEAD`. To discard regenerated changes and restore the tracked versions, use `git restore output/`.

Local execution and smoke tests

Run commands from the repository root. The examples use `uv run python` so they use the pinned environment; `python` also works after activating it. Scripts import `mlye` and helpers from source, and `replicate.py` sets `PYTHONPATH` itself.

```
uv run python replicate.py run --layer 2 --block hetero   # one block
uv run python replicate.py run --only 'ar1n-sweep-*'      # task-id globs
uv run python replicate.py run --workdir /scratch/rep1   # write outputs to a separate tree
```

A task is skipped when its outputs already exist, unless an input was rewritten earlier in the same run. Thus bundles are regenerated whenever their cells are. Use `--force` to rerun tasks regardless. Logs go to `output/logs/<task>.log` inside the chosen working directory.

For an end-to-end smoke test (minutes, with the ICPSR files available):

```
uv run python replicate.py run --smoke --with-bootstrap --layer 1,2 --workdir /tmp/smoke
```

This runs every layer-1/2 task with tiny epoch counts and grids. It exercises code paths, file contracts and exporters, but its numbers are not replication results. Add `--block ar1n,hetero,hockey,timing` to exercise only the simulation blocks without external data.

Cluster execution with scripthut

The following uses scripthut with the configured pythia Slurm backend. Replace `<source>` with your configured source name.

Set `MLYE_STACK_ROOT` to an absolute directory on shared storage, with the same value on every backend using this stack. Configure this outside the repository, in the remote shell environment used by scripthut's SSH sessions and inherited by its Slurm jobs:

```
export MLYE_STACK_ROOT="/path/to/shared/scripthut-stacks"
```

For example, put the export in the remote account's `~/.bashrc` before any early return for non-interactive shells. Exporting it only in the local shell running the scripthut CLI does not configure the remote environment. Installation and task startup both require it; the Python environment lives at `$MLYE_STACK_ROOT/mlye-replication/venv`. If you later change the directory, rerun `scripthut stack install` with `--rebuild` to install it there.

```
scripthut stack install mlye-replication --backend mercury-nb --source <source>   # once; pins from uv.lock
# The stack must also be defined in scripthut.yaml on the source's configured branch.
uv run python replicate.py hut          # writes workflows/hut.replication_<block>[_smoke].json (after changing laye
git add workflows && git commit -m "regenerate workflows" && git push
scripthut source sync <source>
scripthut workflow run workflows/hut.replication_ar1n_smoke.json --source <source> --branch replication --backend py
scripthut workflow run workflows/hut.replication_ar1n.json --source <source> --branch replication --backend pythia-n
# bootstrap: workflows/hut.replication_bpp_with_bootstrap.json instead of workflows/hut.replication_bpp.json
```

The server finds the workflows through the source's `workflows_glob`, a server-side setting: it must match `workflows/hut.*.json` (for example `**/hut.*.json`). `scripthut source view <source>` lists the workflow names exactly as `workflow run` expects them.

Each task runs `python3 replicate.py exec <task-id>` in the server's clone. Tasks in one run share that clone's `output/`, which is how the aggregate and export tasks see the per-cell files. Every output and its manifest are also copied to `$SCRIPTHUT_OUTPUT_DIR`.

For the bpp block, the cluster needs the data. Put the three ICPSR files in `~/data/bpp_ext/`, or a prebuilt `bpp_y_matrix.npy` in `$MLYE_DATA_DIR` (default `~/data`). The prebuilt file is used only if its sha256 matches.

Provenance

Every task writes `output/manifests/<task>.json` with:

- **Code:** git commit and branch, a dirty-tree flag, and the sha256 of the script and of `uv.lock`.
- **Settings:** the exact env and seeds.
- **Hashes:** sha256 and size of every input (upstream outputs, the data matrix, the raw ICPSR files) and every output.
- **Timing:** start and end time and wall clock.
- **Where it ran:** host and Slurm job, partition and node.
- **Runtime, as reported by the script itself:** GPU model, driver, memory and peak use; CUDA, cuDNN, torch and numpy versions; determinism flags.

```
uv run python replicate.py collect [--workdir DIR] [--out DIR] [--strict]
```

This writes `output/PROVENANCE.json` (with `--out DIR`, it also copies the bundles, the layer-3 PDFs and the manifests there; the cluster workflows use this to ship everything with the run). `PROVENANCE.json` traces each bundle and each figure through every upstream task. A figure built from a regenerated bundle chains all the way back through layer 2 to the data; one built from a published bundle lists that file as an external input with its hash. For each task it gives the device, wall time and output hashes. For each bundle it lists the external inputs and the commits used.

It also checks the chain:

- Each input hash a task recorded must equal the output hash recorded by the task that produced that file.
- Each output must still hash to what its manifest says.

A file that was already present and not produced in the run is recorded as `pre-existing`. On the cluster, every workflow ends with a `<block>-collect` task that ships this to the run's outputs.

Refresh the run summary from a run's layer-1/2 manifests with:

```
uv run python replicate.py summary --workdir /scratch/rep1 --readme
```

Omit `--workdir` to use the manifests in this repository. This updates only the generated run-summary block in this README.